Ketamine

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Continuing Education Activity

Ketamine (ketamine hydrochloride) has been approved for general anesthesia either alone or in combination with other medications. It is a superb drug for use in short-term medical procedures that do not require skeletal muscle relaxation, and it has approval for the induction of general anesthesia as a pre-anesthetic to other general anesthetic agents. This activity covers ketamine, including mechanism of action, pharmacology, adverse event profiles, eligible patient populations, contraindications, monitoring, and highlights the role of the interprofessional team in the management of ketamine anesthesia.

Objectives:

- Summarize the mechanism of action of ketamine.
- Identify the indications for initiating therapy with ketamine.
- Outline the adverse event profile of ketamine, along with any possible contraindications to its use.
- Review the importance of collaboration and communication among interprofessional team members to improve outcomes and efficacy for patients receiving ketamine as an anesthetic agent.

Access free multiple choice questions on this topic.

Indications

Ketamine (ketamine hydrochloride) has been approved for general anesthesia either alone or in combination with other medications. It is a superb drug for use in short-term medical procedures that do not require skeletal muscle relaxation. It has approval for the induction of general anesthesia as a pre-anesthetic to other general anesthetic agents.[1][2][3][4]

It also has indications for supplementing low-potency agents such as nitrous oxide. Ketamine is used for short-term procedural sedation, rapid sequence intubation and is very useful in the emergency department setting for these conditions. It is the medication of choice for patients with bronchospasm because of its bronchodilatory properties. It can be utilized for procedures requiring short-term sedation/anesthesia, such as the reduction of fractures and dislocations, as well as in wound repair in uncooperative patients, especially children. It can be safely used in a wide age range, beginning at three months. Because children metabolize ketamine faster than adults, higher dosing is required than in the adult population. Elderly patients, on the other hand, need lower dosing since they are slow metabolizers of this agent.

There is a wealth of evidence indicating the value of ketamine in treating severe pain, including conditions such as trauma, fractures, abdominal and flank pain, low back pain, and extremity pain. When used for pain management, sub-dissociative dosing, otherwise known as low dose ketamine (LDK), is used, either alone or as an adjunct to other pain relief medications. It is safe and effective to use in combination with injectable nonsteroidal pain medications as well as opioids. It has gained greater acceptance as concern has grown regarding opioid use. Clinicians are also using ketamine to treat patients with depression and suicidal ideation successfully. It has been used recreationally as a drug of abuse. The street name for ketamine is "K" or "Special K."

Ketamine is FDA approved for use in anesthesia and procedural sedation. It has been used "off-label" and shown in numerous studies to be safe when used for pain management, as discussed above. Ketamine is not FDA approved for treating depression and suicidal ideation, and such use is off-label. The FDA has recently approved the esketamine (the S enantiomer) form for treatment-resistant depression in conjunction with another oral antidepressant. Recreational use of ketamine is not FDA approved.

At doses below a certain threshold, ketamine produces analgesia and sedation. However, once the critical dosage threshold of roughly 1 to 1.5 mg/kg when given intravenously (IV) or 3 to 4 mg/kg when given intramuscularly (IM) is reached, the characteristic dissociative state abruptly appears. Because of this, the dissociative state is not consistent with the Joint Commission on Accreditation of Healthcare Organizations (JCAHO) definitions of moderate sedation or deep sedation; therefore, ketamine must be considered from a different perspective than drugs with the classical sedation continuum.

Mechanism of Action

Ketamine hydrochloride is a nonbarbiturate dissociative anesthetic. It is a cyclohexanone derivative that is rapidly acting and produces profound anesthesia and analgesia. Its chemical name is ±)-2-(o-chlorophenyl)-2-(methylamino) cyclohexanone hydrochloride; its structural formula is CHClNO. Ketamine is a noncompetitive N-methyl-D-aspartate (NMDA) and glutamate receptor antagonist. It blocks HCN1 receptors. The unique dissociative action and partial agonism on opiate mu-receptors permit the performance of painful procedures in a consistent state of sedation and patient comfort.[5][6][7]

Ketamine's effects in chronic pain, and as an antidepressant, far outlast the actual drug levels and are probably mediated by a secondary increase in structural synaptic connectivity that is mediated by a neuronal response to the ketamine-induced hyper-glutamatergic state." (Sleigh 2014).

The N-methyl-D-aspartate (NMDA) receptor has a significant role in the etiology of depression. Ketamine, through its NMDA antagonistic action, works rapidly in controlling symptoms of depression and acute suicidal ideation. Ketamine may increase glutamate levels and lead to synaptogenesis and elevated levels of brain-derived neurotrophic factor (BDNF).

Ketamine may interact with the sigma receptors. It tends to work by decreasing central sensitization, wind-up phenomenon (development of ongoing, worsening, or chronic pain), and pain memory. Cholinergic, aminergic, and opioid systems appear to play both a positive and negative modulatory role in both sedation and analgesia. Ketamine reverses tolerance to opioids. It is metabolized via the hepatic system by way of N-dealkylation, hydroxylation, conjugation, and dehydration. The half-life of ketamine is approximately 45 minutes.

Ketamine generally maintains normal pharyngeal and laryngeal reflexes and, therefore, permits spontaneous respiration. It slightly enhances or maintains normal skeletal muscle tone and is associated with cardiovascular and respiratory stimulation. These characteristics make it particularly useful in the emergency department setting for short-term procedures, especially as is often the case when a patient has not been "prepped" for an emergency procedure. Since there is no guarantee of maintenance of the pharyngeal and laryngeal reflexes, there can be no assumption that they will "protect" the airway. Additionally, there may be transient minimal respiratory depression if the medication is administered too rapidly or in too high a dose. Therefore, the physician must be ready to perform emergency intubation.

Administration

Ketamine administration can be either IV or IM. The onset of action, when given IV, is within seconds. When administered IM, the onset of action is approximately 4 minutes, with a duration of action from 15 to 30 minutes. Dosing needs to vary according to the desired effect, the patient's age, and underlying conditions. Children metabolize the drug more rapidly than adults and may require higher or additional dosing; elderly patients metabolize ketamine more slowly and may need lower dosing. When using larger total doses of ketamine, administering it more rapidly than normal, or using ketamine with sedatives, barbiturates, or narcotics, exaggerated responses may occur, and prolonged recovery times should be expected.[8][9]

The initial IV dose of ketamine ranges from 1 to 4.5 mg/kg (0.5 to 2 mg/lb) administered over 60 seconds for those 16 years old or older. A dose of 2 mg/kg (1 mg/lb) is the average amount required to produce approximately 5 to 10 minutes of anesthesia/dissociation with the onset of action in about 10 to 30 seconds and a duration of action approximately 5 to 15 minutes. Additional doses can be administered IV or IM to maintain anesthesia without producing significant cumulative effects for more prolonged anesthesia. Either one-half or full additional doses may be repeated as needed. Dosing adjustments will be required if used in combination with other drugs such as IV benzodiazepines or narcotics. When using diazepam in adults, maintenance should be via slow infusion at 0.1 to 0.5 mg/minute of ketamine. Additional doses of diazepam in the amount of 2 to 5 mg are an option.

- In adults, an alternative induction dose is 1 to 2 mg/kg IV at a rate of 0.5 mg/kg/min. IV diazepam 2 to 5 mg can be given over 60 seconds to reduce emergence phenomena. In most cases, 15 mg IV or less will suffice.
- The initial IM dose of ketamine ranges from 6.5 to 13 mg/kg (3 to 6 mg/lb). A 10 mg/kg (5 mg/lb) dose usually produces 12 to 30 minutes of surgical anesthesia with an onset of action in 3 to 5 minutes.
- In children, the IM dosing range is from 9 to 13 mg/kg (4 to 6 mg/lb) and usually produces surgical anesthesia within 3 to 4 minutes following injection.
- Sub-dissociative dosing (low dose ketamine) is dosed at 0.1 to 0.4 mg/kg IV.

Illicit use of ketamine includes snorting or inhalation, and it is ingestable in food/drinks.

Tonic-clonic movements may present during the administration of ketamine. These movements do not mean that additional doses of the anesthetic are required.

Adverse Effects

The most common side effects associated with ketamine are nausea, vomiting, dizziness, diplopia, drowsiness, dysphoria, and confusion. There are reports of emergence phenomenon for approximately 6% to 12% of patients. Rarely, patients experience hallucinations.

A complete list of side effects follows:

- Allergic: anaphylaxis, breathing difficulties, facial, lip, throat, and tongue swelling, hives
- Cardiovascular: arrhythmias, blood pressure, is frequently elevated, bradycardia, hypotension, left ventricular dysfunction in patients with heart failure, respiratory and cardiac arrest
- Gastrointestinal: anorexia, nausea, and vomiting
- Muscular: muscle stiffness and spasms/tonic-clonic movements resembling seizures, enhanced skeletal muscle tone
- Neurologic: confusion, seizures
- Ophthalmologic: diplopia, increased intraocular pressure, nystagmus
- Psychiatric: amnesia, anxiety, confusion, depression, disorientation, dysphoria, dissociative state (patients may not be able to speak or respond purposefully to verbal commands), emergence phenomena/delirium (6% to 12% in different studies and can last for up to 3 hours) including hallucinations, flashbacks, unusual thoughts, extreme fear, excitement, and irrational behavior, insomnia, physical and psychological dependence, addiction when used recreationally. (Drug dependence and tolerance may develop after prolonged use. Withdrawal symptoms may occur if stopping ketamine suddenly.)
- Respiratory: apnea, increased laryngeal, and tracheal secretions, laryngospasm, airway obstruction in infants (may not be drug-related), respiratory depression
- Skin: (infrequently) at the site of injection, local pain, and erythema, morbilliform rash

Emergence delirium may be reduced by decreasing the recommended dose of ketamine, using it in conjunction with benzodiazepines, and/or reducing verbal and tactile stimulation during the drug administration. If "emergence" occurs, a small hypnotic dose of a short-acting barbiturate or benzodiazepine is recommended to terminate the reaction. Diazepam 2 to 5 mg IV push given over 1 minute (15 mg or less in most cases) can reduce emergence phenomena. "Emergence" is not typical in children younger than 15 years or elderly patients greater than 65 years. In a randomized controlled study of adult emergency department patients, Sener et al. found a significant reduction in the incidence of agitation when coadministering midazolam with ketamine for procedural sedation.

Contraindications

Ketamine is contraindicated in patients with underlying conditions in which increased blood pressure would pose a risk of complications such as aortic dissection, uncontrolled hypertension, myocardial infarction, or aneurysms.

- It is contraindicated in those who have shown prior hypersensitivity to the drug.
- It is not recommended for use during obstetrics, pregnancy, or breastfeeding as it is unknown if this medication passes into breast milk.
- Care must be used in patients who are intoxicated with ethanol due to additive sedation.
- It is contraindicated in patients with schizophrenia due to the potential for exacerbating the underlying condition.

In patients with cerebrospinal fluid (CSF) elevations, the use of ketamine is controversial due to questionable elevations of the CSF caused by ketamine. Some studies indicate that the concern for CSF elevation with ketamine has been overstated. Kropf et al. found similar hemodynamic properties compared with etomidate. Newer research suggests that ketamine may improve cerebral perfusion pressure and may, in fact, have neuroprotective properties.

Monitoring

Monitor vital signs and cardiac function. Blood pressure, pulse, breathing, and oxygen saturation always need to be monitored when using ketamine.

Be prepared for intubation when administering ketamine.

Neuropsychiatric function always requires monitoring, and a patient must be at baseline before discharge from care. Even after a patient's return to baseline, the patient discharge should be under the supervision of another responsible adult. Patients should not drive, use heavy machinery, or perform other potentially hazardous activities for up to 24 hours following ketamine administration.

Toxicity

Ketamine is potentially fatal in alcohol-intoxicated patients.

*May increase CSF pressure (controversial-see above/contraindications)

Enhancing Healthcare Team Outcomes

Ketamine's primary use is by anesthesiologists, anesthesia nurses, dentists, emergency department physicians, pain specialists, and psychiatrists; these various disciplines need to function as an interprofessional team to optimize the drug's effectiveness while minimizing the potential for adverse events. [Level 5] It is an excellent drug for analgesia and sedation. At low doses, it does not produce a dissociative state. All healthcare workers who use ketamine should know the indications and contraindications. Nurses and pharmacists can play a crucial role during ordering and administration, guarding against improper dosing and checking for adverse events and drug interactions. Also, monitoring of the patient's vital signs is mandatory while infusing the drug. One should always have resuscitative equipment in the room in case intubation is required. Before discharge, the clinical team must record the patient's neuro-vitals. Patients should not drive, use heavy machinery, or perform other potentially hazardous activities for 24 hours after the administration of ketamine.